Electromagnetic induction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain why changing magnetic flux can induce an emf.

SUCCESS CRITERIA - I can...

1. Identify a change in magnetic flux.
2. Predict whether a stationary arrangement produces sustained induction.
3. Use Lenz's law to describe opposition to change.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use "induction" and "flux" to explain your model or result. Sentence starters are optional.

induction: producing an emf through a changing magnetic flux

flux: a measure of magnetic field passing through a chosen area

emf: energy supplied per charge, measured in volts

01 Bring back an earlier idea

A 3 W device runs for 5 s.
Find transferred energy.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $\Phi = BA$; $|emf_{avg}| = N|\Delta\Phi|/\Delta t$.
Uniform perpendicular field for $\Phi = BA$. N is coil turns; the induced effect opposes the flux change.

Worked reasoning: A changing magnetic flux through a coil can induce voltage. Holding a magnet stationary relative to a fixed coil gives no induction from relative motion in this simple setup.

Your turn: Draw a magnet approaching a coil and mark the changing field through the loop. Contrast it with a stationary case.

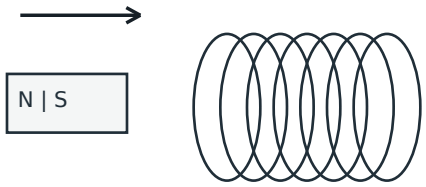
Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Look and explain

Compare a magnet held still beside a fixed coil with a magnet moving into the coil. Which can produce a changing flux?

Sentence starter: Both ... . They differ in ... because ... .



Coil

Marzano check - circle: ① ② ③

04 Build the solution

In a supplied trial, moving a magnet into a coil gives a positive meter deflection.
Predict the sign when the same magnet is pulled out along the same path.

Sentence starter: I predict ... . My model supports this because ... .

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Use the idea

Why does Lenz’s law oppose the change in flux rather than always oppose the field itself?

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

06 Choose a method

Keep charge separation fixed and multiply one charge magnitude by 4.
Compare electric-force magnitude.

Word help: separation = distance between the specified objects or centers

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

07 Exit check

Name two ways to increase the magnitude of induced emf in a simple magnet-and-coil trial.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Predict how faster approach and more turns can change induced voltage.
State which comparisons must keep the other conditions fixed.

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: 1 2 3